

Haocheng He

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PROFESSIONAL SUMMARY

Industrial Engineering graduate from GDUT with a Bachelor of Laws minor degree, currently pursuing an MEng in Aviation Engineering and Operation Management at PolyU. Completed 4 internships across AI hardware integration, airport emergency simulation, manufacturing improvement, and assembly-line optimization, plus a time-series RUL prediction project. Technically curious, fast-learning, and comfortable with hands-on engineering, data modeling, and high-intensity execution.

EDUCATION

The Hong Kong Polytechnic University Hong Kong SAR, China
Master of Engineering in Aviation Engineering and Operation Management Sep 2025 – Jan 2027
GPA: 3.95 / 4.3 (Current)
Thesis: Simulation Modelling and Optimization for Airport Ferry Vehicle Scheduling under Uncertainty

Guangdong University of Technology Guangzhou, China
Bachelor of Engineering in Industrial Engineering Sep 2021 – Jun 2025
GPA: 3.67 (Top 20%)
Thesis: AnyLogic-Based Simulation and Optimization of Emergency Special Vehicle Routing at a Large Airport
Honors: Excellent Graduation Thesis + Innovation Award

Bachelor of Laws (Minor Degree in Law) Sep 2022 – Jun 2025
GPA: 3.61
Thesis: Coordination Between Copyright Protection and Privacy Protection in Digital Rights Management

EXPERIENCE

AI Engineer Assistant May 2025 – Apr 2026
Shenzhen Wuxian Hongyin Technology Co., Ltd. Shenzhen, China

- Participated in smart piano R&D based on Huawei Ascend 310B, covering PCB hardware tests (CPLD/STM32), piano motor tests, full-bridge motor driver schematic/layout, and STM32 firmware development.
- Supported partial module integration for a camera- and Ascend-based piano fingering recognition system; reviewed code, understood system architecture, and helped locate issues.
- Maintained Ubuntu servers and Huawei switches/routers; also supported procurement, administration, and technical interviews.

Research Assistant Sep 2023 – Jun 2025
Guangdong University of Technology (with Guangzhou Baiyun Int'l Airport) Guangzhou, China

- Built an airport road-network, apron, and runway-area simulation model in AnyLogic, Java, and MySQL for aircraft landing and emergency fire-truck dispatch scenarios.
- Designed single/multi-vehicle, single/multi-event, and runway-road-closure scenarios; compared D* Lite, A*, Dijkstra, BFS, and GA routing algorithms.
- Completed 4 emergency-scenario comparisons across 5 routing algorithms; D* Lite achieved the highest average reliability index (1.342).

Industrial Engineering Intern Apr 2024 – May 2024
Midea Group — Kitchen & Water Heater Division Foshan, China

- Conducted stopwatch time studies across all workstations on a dishwasher sheet-metal line, built operator workload tables, and identified 5 over-cycle steps.
- Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.

- Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis.
- Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation. Discussed findings with line engineers.

PROJECTS

Aircraft Engine Remaining Useful Life Prediction (Course Project)

Python / C-MAPSS

- Built an RUL prediction workflow on NASA C-MAPSS FD001 multivariate time-series data; cleaned operating-condition data and 21 sensor signals, constructed sequence features, and applied PCA.
- Compared LSTM, BiLSTM, Transformer, and Random Forest baselines for time-series forecasting; after iterative tuning, the BiLSTM model reached about 8% final error from an initial model error of about 50%.

SKILLS

Programming & Data: Python, Java, SQL/MySQL, C/STM32, JavaScript/TypeScript

AI & Automation: Claude Code, Codex, Hermes, ChatGPT, Prompt Engineering, Harness Loop Engineering, AI Agent workflows, model deployment/evaluation

Machine Learning: Random Forest, LSTM, BiLSTM, Transformer, PCA, model training/evaluation

Simulation & Optimization: AnyLogic, Plant Simulation, OR-Tools, Dijkstra, A*, BFS, genetic algorithms, line balancing, ECRS

Engineering Tools: Ubuntu/Linux, Git, LCEDA, oscilloscope, logic analyzer, Huawei switches/routers

Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational)

PUBLICATIONS & INTELLECTUAL PROPERTY

Invention Patents

- PI Coating Machine Motion Control via MCMC Sampling — Zhou Xianzhong, He Haocheng, He Zhiyuan (2026). CN122085794A, Guangdong University of Technology.
- Coating Control Method and System (Deep Reinforcement Learning) — Zhou Xianzhong, He Haocheng, He Zhiyuan (2025). CN121325552A, Guangdong University of Technology.
- Home Solar Energy Storage System — Adaptive Energy Regulation — Zhou Xianzhong, Zeng Huangtao, Yu Daming, He Haocheng (2024). CN117767437A, Guangdong University of Technology.

Utility Model

- Integrated Wire Harness Production Equipment — Zhao Xiang, He Haocheng, Hu Bofeng. CN221352455U.

Software Copyright

- Enterprise Financial Data Analysis Software V1.0 — Li Yisi, He Haocheng (2024). Reg. No. 2024SR0146827.
- Urban Flood Intelligent Mobile Emergency Command Dispatch System V1.0 — Lin Chengfeng, Li Zeyin, Yang Hua, He Haocheng (2024). Reg. No. 2024SR0549247.